

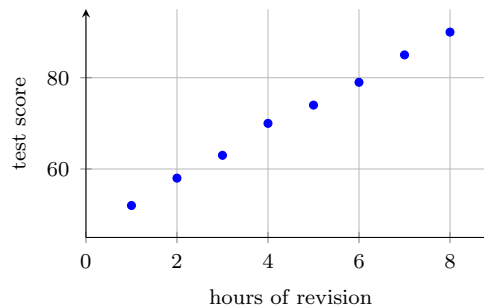
Direction, Form, and Strength in Scatter Plots

Reading a scatter plot, then naming its direction, its form, and how strong it is

Directions

- **Direction:** positive (both go up together), negative (one goes up as the other goes down), or none.
- **Form:** linear if a single straight line would follow the points, non-linear if the pattern bends.
- **Strength:** strong if the points hug the pattern closely, weak if they are spread widely around it.
- Answer the number question first, then write the description in full sentences.

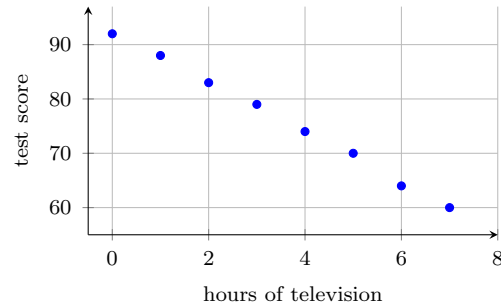
1. Each point shows one student's revision time and test score.



- (a) The student who revised for 5 hours scored _____

- (b) Describe the direction of the association.

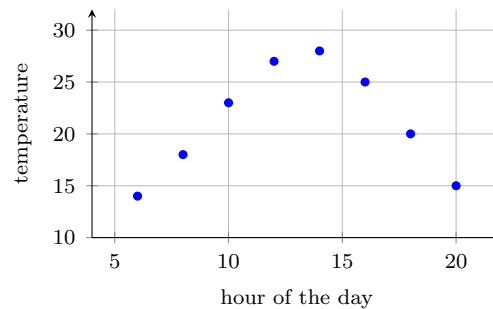
2. Each point shows one student's daily television time and test score.



- (a) The highest score plotted is _____

- (b) Describe the direction of the association.

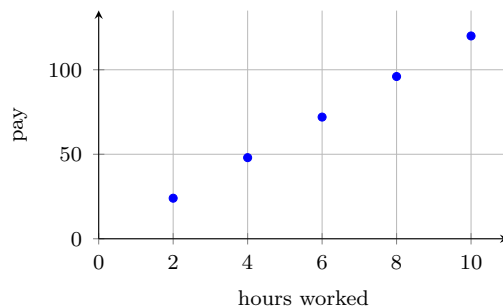
3. Each point shows the temperature in degrees Celsius at one hour of a summer day.



- (a) The warmest reading is at hour _____

- (b) Describe the form of the association, and say why a single straight line would not fit it.

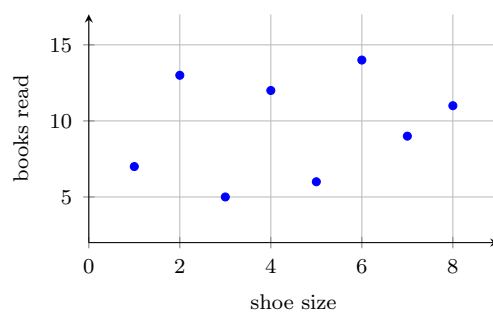
4. Each point shows hours worked and pay earned, in dollars, for one shift.



- (a) The mean of the five pay amounts is _____

- (b) Describe the strength of the association.

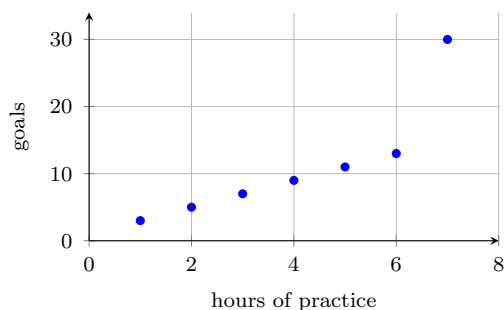
5. Each point shows one student's shoe size and the number of books they read last month.



- (a) The range of the numbers of books read is _____

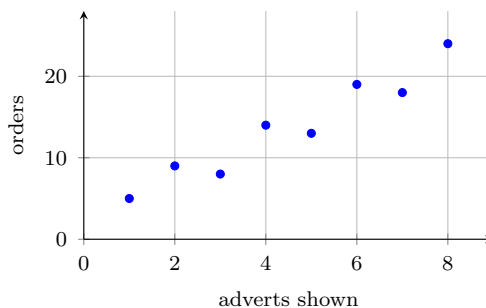
- (b) Describe the direction of the association.

6. Each point shows hours of practice and the number of goals scored by one player.



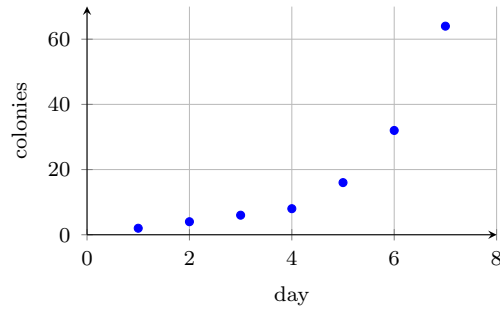
- (a) The greatest number of goals plotted is _____
- (b) Circle the point that does not follow the pattern, and say what makes it unusual.

7. Each point shows the number of adverts shown and the number of orders taken that day.



- (a) The median number of orders is _____
- (b) Describe both the direction and the strength of the association.

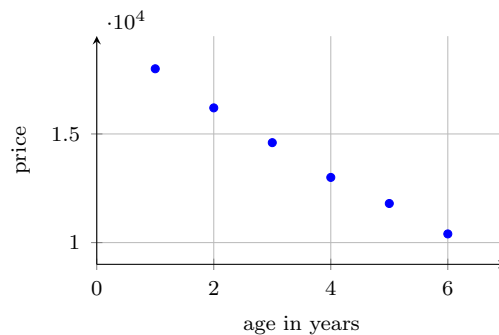
8. Each point shows the day of an experiment and the number of bacteria colonies counted.



- (a) Day 7 minus day 1 gives a difference of _____

- (b) Describe the form of the association.

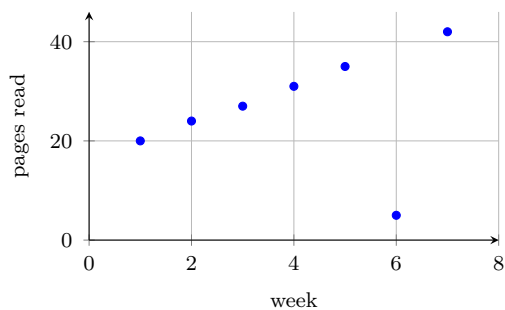
9. Each point shows the age of a used car in years and its price in dollars.



- (a) The mean price of the six cars is _____

- (b) Describe the direction, the form and the strength of this association in one or two sentences.

10. Each point shows the week of a reading challenge and the number of pages one student read.



(a) In the week that breaks the pattern the student read _____pages

(b) Describe the association shown by the other six points, and explain how the odd week changes the impression the plot gives.
